

EEG for LLMs: A Telemetry Layer for Online Uncertainty Monitoring and Decision Policies

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Abstract

Token streams are a human-oriented interface that can obscure generation dynamics and encourage brittle analyses (e.g., relying on chain-of-thought text). We introduce an “EEG-like” telemetry layer for autoregressive decoding that records lightweight internal signals during generation—uncertainty, surprisal, distribution shift, and sparse layer summaries—yielding real-time traces of model state evolution *without* parsing chain-of-thought text. Across three model families and three task types (27 runs = 3 models $\times$ 3 tasks $\times$ 3 seeds), we find that telemetry signatures vary strongly across models and tasks, and that early-window uncertainty can predict failures above random on labeled tasks (entropy AUC 0.61–0.74 on GSM8K and 0.35–0.75 on TriviaQA). As an application demo, we show how telemetry can gate simple downstream policies (accept / retry / route) on a Llama-8B $\rightarrow$ Qwen-32B (4-bit) pair, improving accuracy by up to +10.5 points on GSM8K (route-only; $1.42\times$ cost proxy) and +1.5 points on TriviaQA (cascade; $2.81\times$ cost proxy). We release a reproducible pipeline, canonical benchmarks, and visualization tools.¹ We view this work as a first step toward systems that progressively reduce reliance on human token interfaces.

1 Introduction

Large language models (LLMs) are typically observed through their token outputs. This is a convenient human interface, but it is also slow and lossy: a token stream compresses a high-dimensional internal computation into a one-dimensional sequence, and it tempts practitioners to overfit to surface text patterns such as chain-of-thought (CoT) strings. This paper asks a simple question: *can we treat generation as an online dynamical process and instrument it with a lightweight telemetry layer*, analogous to an “EEG” for decoding?

We focus on telemetry that is (i) **online** (available while decoding), (ii) **lightweight** (scalar or low-dimensional per token), and (iii) **CoT-agnostic** (does not parse intermediate reasoning text). The goal is not interpretability-by-default, but *observability*: exposing signals that allow us to quantify uncertainty dynamics, compare model behaviors across tasks, and drive simple decision policies such as retrying under alternative decoding, routing to an expert, or early stopping.

Context and scope. CoT prompting can improve reasoning accuracy [23, 22], but intermediate reasoning strings are also an unreliable interface for monitoring: they are long, hard to standardize, and can be misleading when treated as literal explanations. Our focus is complementary: rather than

¹Code and tooling: <https://github.com/nick-yudin/grokking-research> (folder: `latent_configuration/01_interpretable_latent_probes`). Canonical paper bundles and figures will be mirrored at https://github.com/nick-yudin/Generalization/tree/main/papers/eeg_for_llms.

extracting structure from text, we instrument the decoding process itself and ask what can be done with low-bandwidth internal signals. We intentionally start with simple, model-agnostic features (entropy/surprisal/distribution drift) and reserve learned probes and latent-control mechanisms for follow-up work.

Contributions.

- **Telemetry interface.** We define an “EEG-like” telemetry layer for autoregressive decoding that logs uncertainty and instability signals per token (entropy, surprisal, ΔL_1 distribution shift) together with sparse layer summaries.
- **Canonical benchmark & reproducible evaluation.** We release a canonical dataset of 27 monitoring-only runs (3 models $\times$ 3 tasks $\times$ 3 seeds) and an end-to-end tooling pipeline (collection, analysis, visualization, and report-pack generation).
- **Early predictiveness.** On labeled tasks, we show that early-window uncertainty predicts failures above random, with entropy AUC ranging 0.61–0.74 on GSM8K and 0.35–0.75 on TriviaQA (task/model dependent).
- **Actionable demo.** We demonstrate telemetry-gated downstream policies (retry/routing variants) on a Llama-8B $\rightarrow$ Qwen-32B (4-bit) pair, showing measurable accuracy gains without parsing CoT text.

2 Telemetry Layer

2.1 Signals

At each generated token step t , we compute a small set of telemetry signals from the model’s next-token distribution restricted to its top- k logits (we use $k = 50$ in canonical runs). Let $p_t^{(k)}$ denote the normalized top- k probability vector. We log:

- **Top- k entropy** $H_t = -\sum_i p_{t,i}^{(k)} \log p_{t,i}^{(k)}$ and a normalized variant $H_t / \log k$.
- **Chosen-token surprisal** $S_t = -\log p(\hat{y}_t)$, where $\hat{y}_t$ is the chosen token under the decoding policy.
- **Distribution shift** $\Delta_t = \|p_t^{(k)} - p_{t-1}^{(k)}\|_1$ as a proxy for “hypothesis churn”. We align the truncated distributions by *token ID* over the union support $\text{top-}k(t) \cup \text{top-}k(t-1)$ (missing probabilities are treated as zero).
- **Repetition proxies** such as the maximum streak of repeated chosen-token IDs.
- **Sparse layer summaries** (e.g., hidden-state norms) for a small set of layers (default: last 4 layers).

The key design choice is to keep telemetry *low-bandwidth* enough to be collected online with minimal overhead, while remaining informative enough to support error prediction and downstream decisions.

2.2 Telemetry windows

Many downstream decisions require *early* signals. We therefore compute windowed features over the beginning of each completion (e.g., the first 16 generated tokens). We also compute “early-risk” AUC at multiple horizons (first 8/16/32/64/128 tokens) to measure how quickly predictiveness emerges.

2.3 Early-risk AUC

For labeled tasks (GSM8K and TriviaQA), we evaluate how well a telemetry score z predicts failure using the area under the ROC curve (AUC). We treat each example as *incorrect* if the produced final answer is wrong *or* the output violates a strict format contract (when enabled). AUC is computed using early-window statistics (e.g., mean normalized entropy over the first 16 tokens) and, separately, as a function of horizon N (AUC@8, AUC@16, ...) using the first N tokens of telemetry. An AUC of 0.5 corresponds to random ranking; values below 0.5 indicate *polarity inversion* (the score is predictive in the opposite direction), in which case negating the score yields an equivalent AUC of $1 - \text{AUC}$.

2.4 System overview

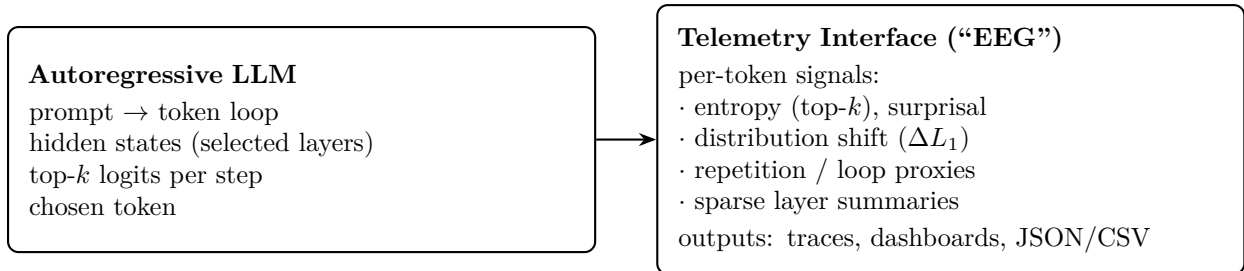


Figure 1: System overview: a telemetry interface around an autoregressive decoder enables downstream decisions without chain-of-thought parsing.

3 Experimental Setup

3.1 Models

We evaluate three instruct-tuned model families:

- Qwen/Qwen2.5-14B-Instruct
- meta-llama/Meta-Llama-3.1-8B-Instruct
- mistralai/Mistral-7B-Instruct-v0.3

We keep decoding deterministic ($T = 0$) to reduce sampling variance and to isolate model/task effects in telemetry.

3.2 Tasks, prompts, and labels

We use three task types. In all settings we collect 200 samples per run and repeat each configuration across 3 random seeds.

Table 1: Canonical Benchmark & Reproducible Evaluation (mean over 3 seeds). Primary metric: lenient correctness. Interface compliance: strict format error rate. Early-risk AUC uses an early-window entropy/surprisal signal. “Tokens” denotes the mean number of generated tokens per completion (proxy cost).

Task	Model	Lenient acc	Format err	AUC(ent)	AUC(spr)	Tokens
gsm8k	Qwen2.5-14B-Instruct	0.80 ± 0.01	0.04 ± 0.02	0.61	0.61	52.50 ± 0.71
gsm8k	Meta-Llama-3.1-8B-Instruct	0.57 ± 0.03	0.15 ± 0.01	0.67	0.67	48.60 ± 0.23
gsm8k	Mistral-7B-Instruct-v0.3	0.25 ± 0.03	0.90 ± 0.01	0.74	0.71	59.64 ± 0.66
trivia	Qwen2.5-14B-Instruct	0.42 ± 0.02	0.51 ± 0.02	0.75	0.73	15.78 ± 0.13
trivia	Meta-Llama-3.1-8B-Instruct	0.32 ± 0.01	0.70 ± 0.04	0.71	0.59	23.85 ± 1.14
trivia	Mistral-7B-Instruct-v0.3	0.52 ± 0.02	0.90 ± 0.01	0.35	0.32	25.57 ± 0.30

- **GSM8K** (math; gold numeric answer) [3]. We prompt models to output two lines: **WORK:** `<short reasoning>` and **FINAL:** `<number>`. We evaluate numeric correctness after parsing.
- **TriviaQA** (short factual QA; gold answer + aliases) [9]. We prompt a strict single-line **FINAL:** `<answer>` output for reliable parsing, and evaluate using a lenient alias match.
- **Wikipedia continuation** (no gold labels; open-ended generation). We sample Wikipedia articles via the Hugging Face Datasets interface [11] and generate continuations; we report telemetry distributions and monitoring-only stop behavior.

3.3 Primary vs secondary metrics

We report two axes separately: (i) **answer quality** (primary; lenient semantic correctness on labeled tasks), and (ii) **interface compliance** (secondary; strict format error rate under the chosen output contract). This separation is important in practice: some models achieve reasonable semantic correctness while frequently violating strict output formatting.

4 Results: Monitoring Only

4.1 Canonical Benchmark & Reproducible Evaluation

4.2 Model and task variability

Figure 2 summarizes lenient accuracy on labeled tasks; Figure 3 reports strict format error rates. These plots highlight that telemetry must be interpreted in context: model families differ substantially in both correctness and interface compliance, and these differences interact with task type.

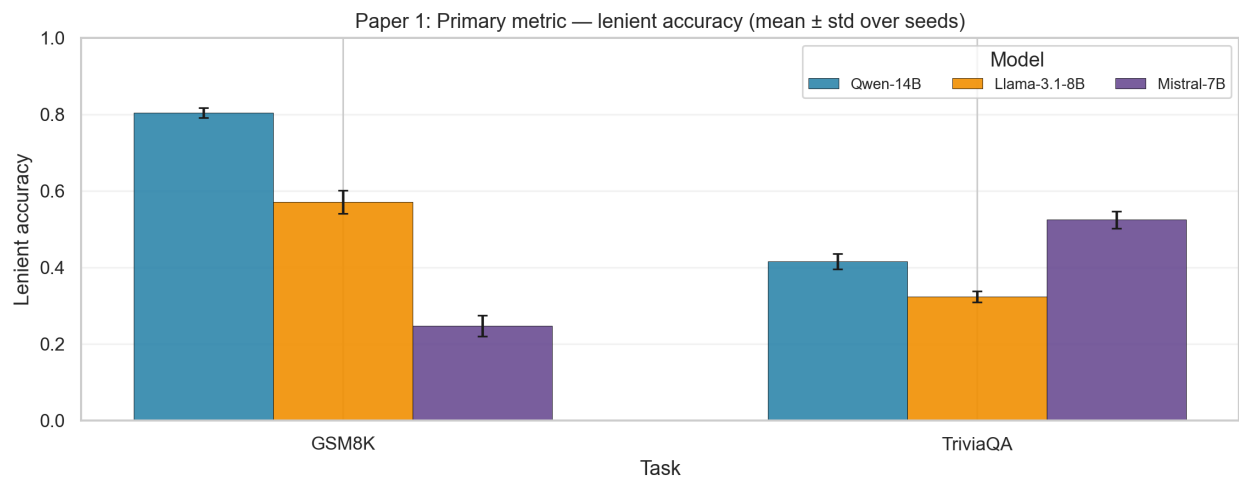


Figure 2: Primary metric for labeled tasks: lenient semantic correctness (mean over seeds).

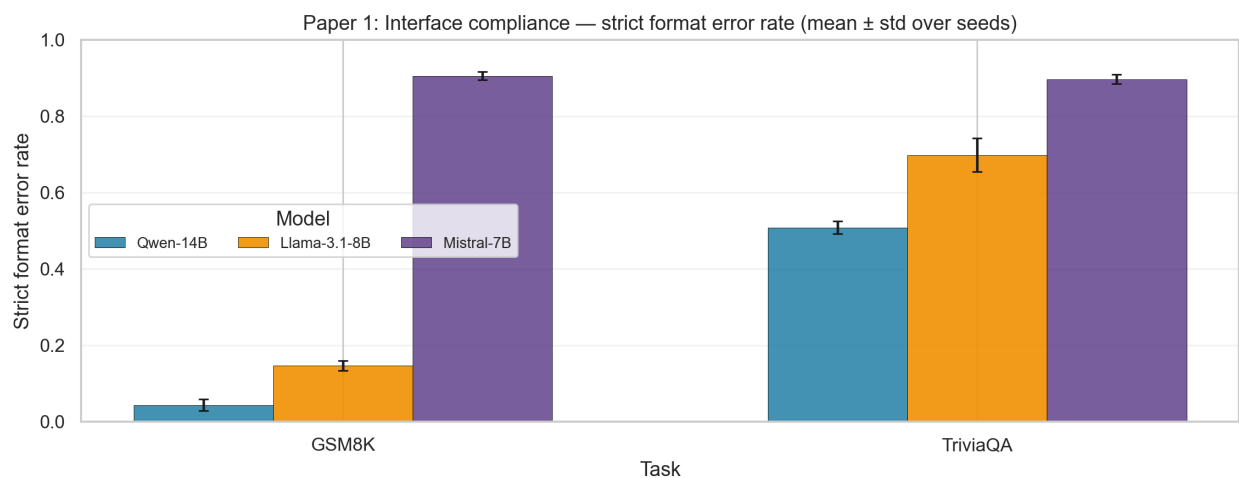


Figure 3: Interface compliance: strict format error rate (mean over seeds).

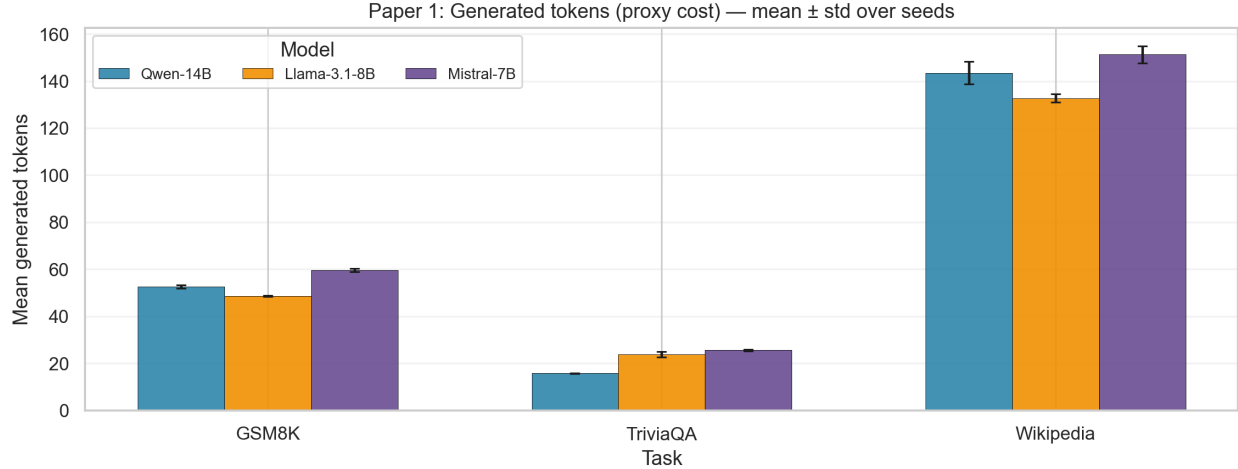


Figure 4: Mean generated tokens as a proxy for cost (mean over seeds).

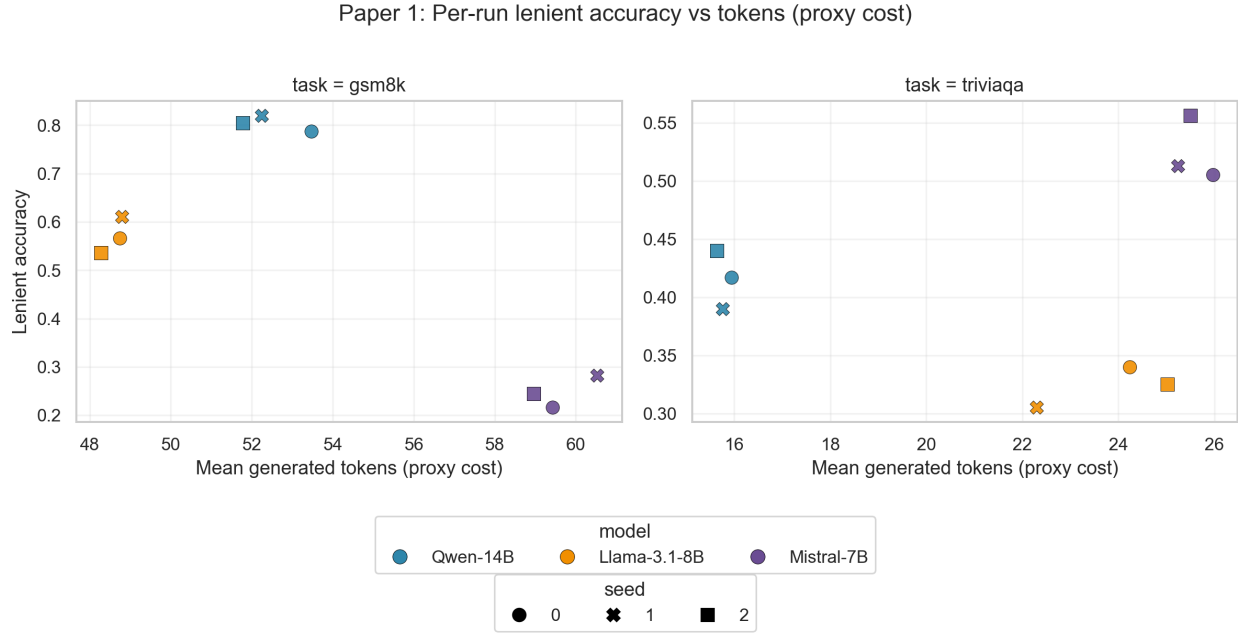


Figure 5: Per-run lenient accuracy vs. mean generated tokens (proxy cost), split by labeled task.

4.3 Early predictiveness of failures

Figure 6 summarizes early-window AUC for entropy and surprisal signals. We observe substantial task/model dependence: for example, Mistral exhibits high AUC on GSM8K while underperforming on TriviaQA, despite high strict format error rates on both tasks. This supports the view that telemetry captures *model dynamics* rather than reflecting only final accuracy. In a few settings AUC falls below 0.5 (polarity inversion), indicating that the raw score is predictive in the opposite direction; in practice this is handled by calibrating the score polarity per model/task (or by using $1 - \text{AUC}$ for a sign-flipped score). We also split failure labels into (i) *format failures* and (ii) *semantic errors conditioned on format correctness*; this clarifies that some telemetry signatures primarily

detect interface non-compliance rather than semantic mistakes (Appendix A).

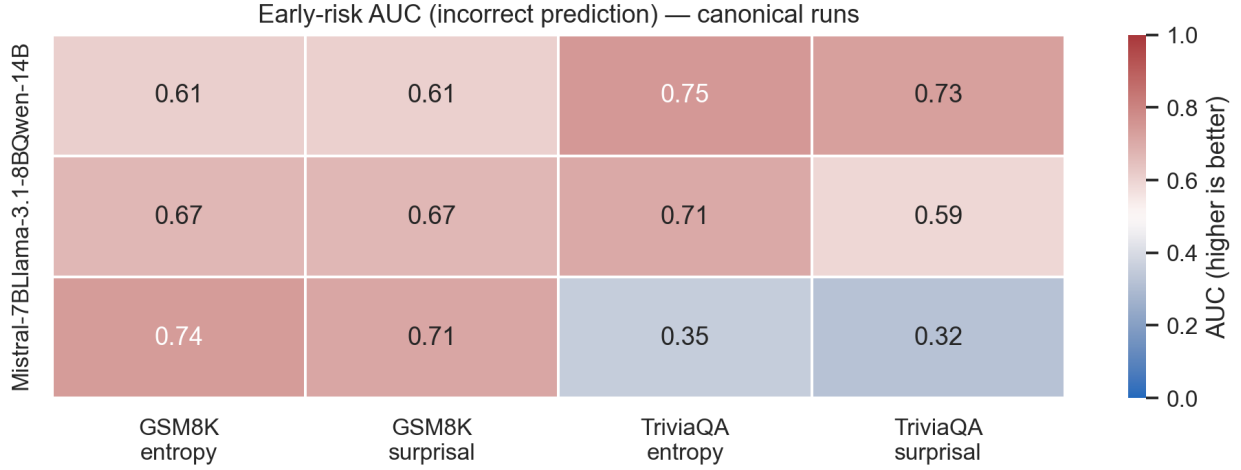


Figure 6: Early-risk AUC (predicting incorrect outputs) by model and task for entropy and surprisal signals.

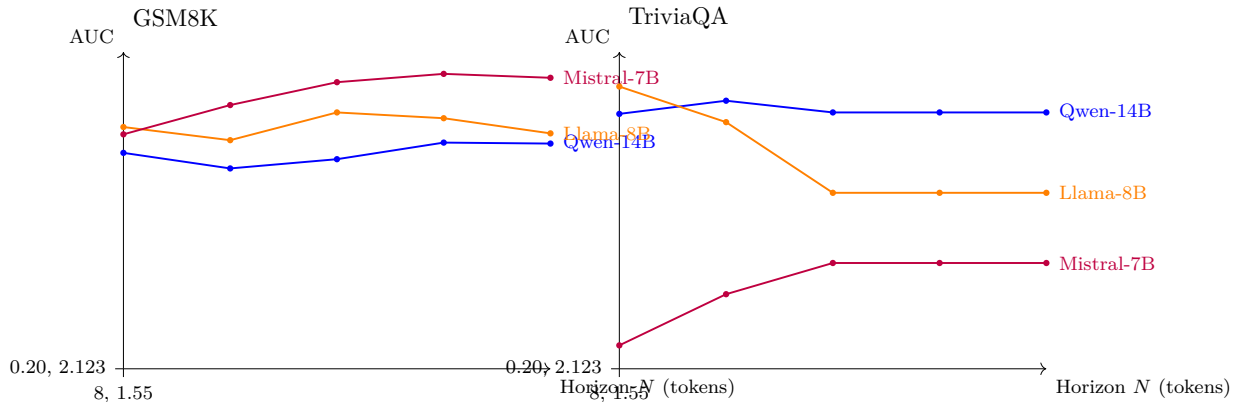


Figure 7: AUC vs. horizon N for mean normalized entropy using the first N tokens of telemetry (mean over seeds). In many settings, much of the predictive signal is available within the first few dozen tokens.

4.4 Wikipedia continuation (no gold labels)

Wikipedia continuation provides a stress test beyond labeled QA. We avoid factuality claims and instead report telemetry distributions, generation lengths, and monitoring-only stop behavior. In our canonical runs (200 samples per run), monitoring-only uncertainty stops trigger in 108–131 cases depending on model (54–66%), indicating that early uncertainty dynamics vary substantially across model families. This suggests telemetry captures model-level confidence differences even without gold labels, and can support monitoring/guardrails in open-ended generation.

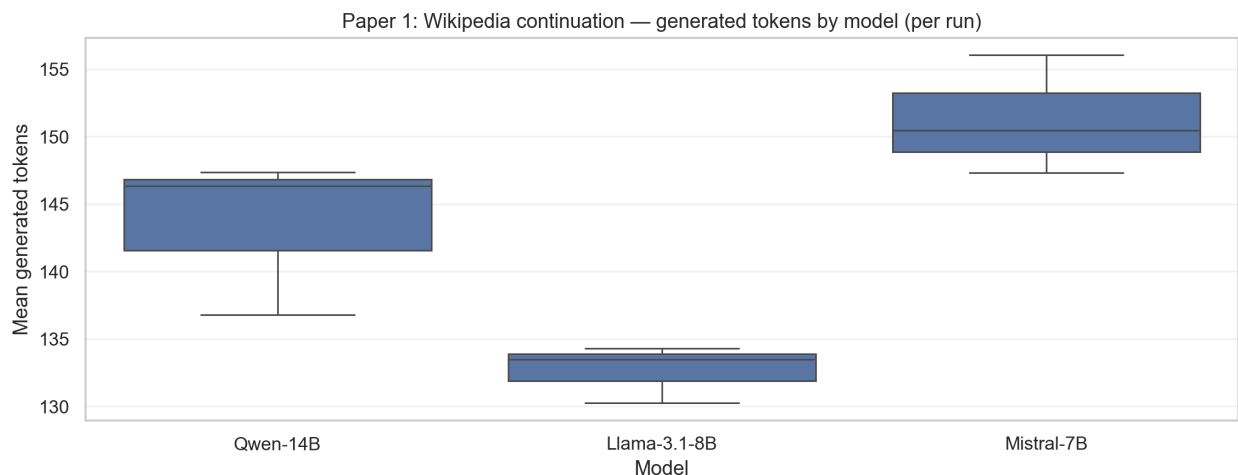


Figure 8: Wikipedia continuation: generated-token count distribution by model (per run).

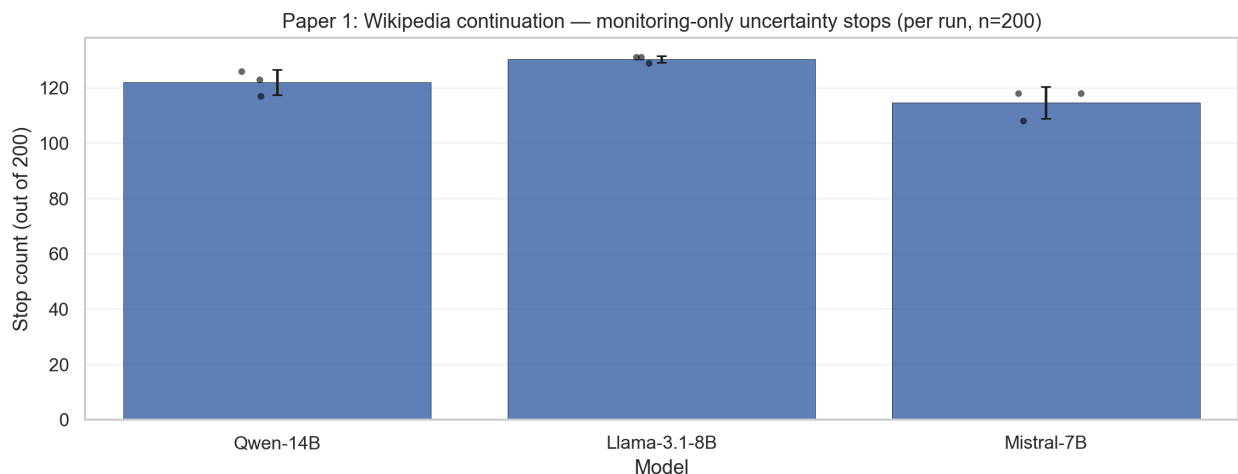


Figure 9: Wikipedia continuation: monitoring-only uncertainty stops (per run). Counts are out of $n = 200$ samples per run.

4.5 Case study: a single “EEG” trace

To provide intuition, Figure 10 shows a representative trace. Such traces can be used for qualitative debugging (“where did uncertainty spike?”) and as an input substrate for downstream policies (retry/route/stop) without inspecting CoT text.

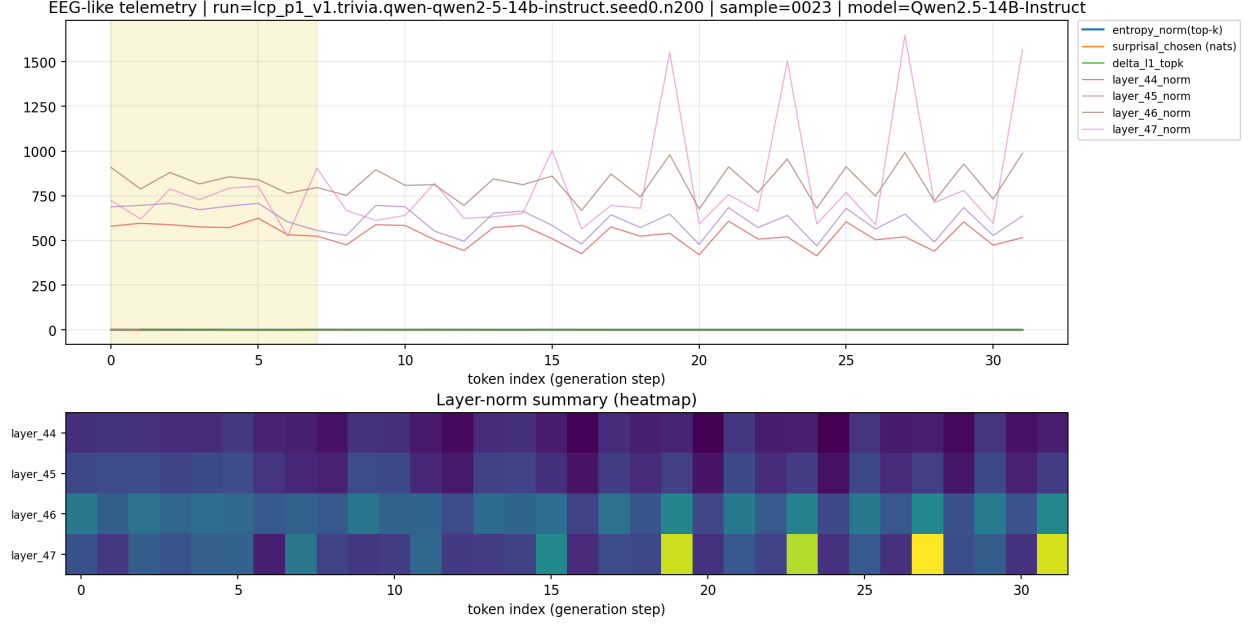


Figure 10: Hero EEG trace example (TriviaQA; Qwen-14B; seed 0; sample 0023). The shaded region marks the early window used for AUC features (e.g., $N = 8$) even though the full trace is shown.

5 Downstream Policy Demo: Telemetry-Gated Cascade

Monitoring is only useful if it supports action. As an application demo (not an optimized router), we use telemetry as a gating signal for a simple cascade policy: attempt a cheaper base model first, optionally retry under alternative decoding when risk is moderate, and route to a stronger expert when risk is high.

Policy sketch. Let r be a telemetry-derived risk score computed online. A two-threshold cascade can be expressed as: (i) accept if $r < \theta_{\text{retry}}$, (ii) retry if $\theta_{\text{retry}} \leq r < \theta_{\text{route}}$, (iii) route to an expert model if $r \geq \theta_{\text{route}}$. We use mean normalized entropy over the first 8 tokens as r in canonical cascade runs.

Evaluation protocol. We run a Llama-8B base model and log retry candidates and expert predictions (Qwen-32B, 4-bit) for routed samples. Using these logs, we can compute counterfactual “retry-only” and “route-only” policies without re-running models. We report intervention rate (fraction of samples that trigger retry and/or routing) and a simple cost proxy: base calls cost 1.0, retry cost adds extra base calls, and expert calls are weighted by a constant multiplier (used only for illustrative trade-off plots). In canonical runs we use entropy-based risk with $(\theta_{\text{retry}}, \theta_{\text{route}}) = (0.35, 0.40)$ for GSM8K and $(0.40, 0.50)$ for TriviaQA, with a simple consensus gate on whether to trigger routing. We emphasize that this section is an *application demo*: we do not optimize the router for ROI, and the best operating point may be “route-only” (skipping retries) depending on the task and expert quality.

Table 2: Telemetry-driven downstream policy demo (Llama-8B base, Qwen-32B expert in 4-bit). Primary metric is lenient semantic correctness; strict format errors are reported separately in the monitoring benchmark (Table 1). “Cost” is a simple proxy combining base retries and expert calls.

	Task	Policy	Acc	Δ Acc	Intervene	Cost
2	gsm8k	none	0.592	+0.000	0.0%	1.00
	gsm8k	retry	0.589	-0.003	40.3%	3.02
	gsm8k	route	0.697	+0.105	23.9%	1.42
	gsm8k	cascade	0.692	+0.101	40.3%	3.43
	trivia	none	0.355	+0.000	0.0%	1.00
	trivia	retry	0.362	+0.007	32.5%	2.62
	trivia	route	0.362	+0.007	10.9%	1.19
	trivia	cascade	0.370	+0.015	32.5%	2.81

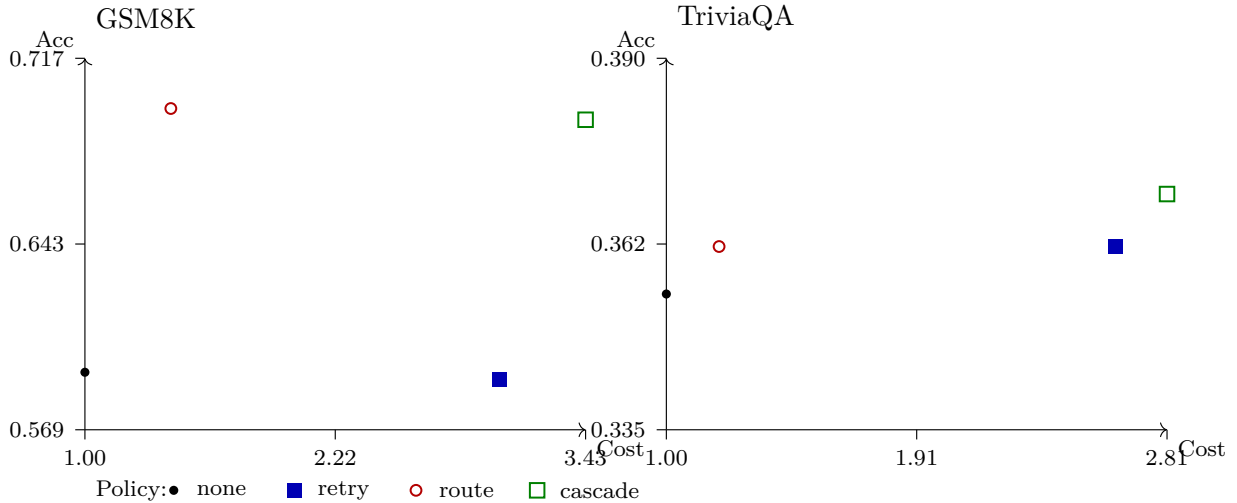


Figure 11: Policy trade-offs computed from the same canonical cascade runs without re-running models (counterfactual “retry-only” and “route-only” reuse logged retry candidates and expert predictions).

6 Related Work

Uncertainty, calibration, and selective prediction. Our telemetry layer focuses on online uncertainty signals (entropy, surprisal) and their dynamics. This connects to uncertainty quantification and selective prediction / abstention, including conformal prediction and risk-controlled refusal / calibration frameworks (see, e.g., [20, 13, 14, 1, 16, 24]).

Routing and cascades. Telemetry-gated retry and routing is closely related to model cascades and LLM routers [4, 15, 12]. Our cascade demo is intentionally simple and serves as an application of monitoring rather than an optimized router; stronger routers typically use learned quality estimators,

²Note: the monitoring benchmark in Table 1 uses a canonical $n = 200$ subset (mean over 3 seeds), while this policy demo uses larger evaluation sets (GSM8K test $n = 1319$, TriviaQA validation $n = 1000$; seed0) to reduce variance. This can shift absolute accuracies even when the model/prompt template is unchanged.

preference-trained selectors, or compute-aware objectives. Multi-sample strategies can also be viewed as a form of “retry” and selection. Confidence-informed self-consistency and self-certainty can improve best-of- N selection, and representation-consistency signals can help aggregate candidates without a heavy verifier [19, 10, 7].

Internal-state monitoring and interpretability. We do not attempt to produce mechanistic interpretations of hidden states. Instead, we treat decoding as an observable process and expose lightweight signals for monitoring and control. This is complementary to interpretability and feature-discovery lines such as probing and latent-state monitoring [5, 13]. Learned probes on internal activations can achieve substantially higher failure-prediction performance than our zero-shot telemetry, but introduce training data requirements and generalization risks [6]. Probing-based detectors can also fail to generalize under distribution shift [21], and several recent works suggest that intermediate-layer signals may be more predictive for certain failure modes such as hallucination [18, 2].

Toward latent-only reasoning and communication. Follow-up papers in this series aim to reduce dependence on human token streams. Recent work explores compressing CoT into continuous spaces or enabling “silent” reasoning [17, 8]. We view Paper 1 as establishing the instrumentation substrate needed to evaluate these directions rigorously.

7 Limitations

Continuation tasks have no gold labels; we report monitoring-only statistics and avoid factuality claims. Some models (notably Mistral) violate strict output contracts at high rates; we treat this as interface compliance and report it separately from answer correctness. Our telemetry features are intentionally simple and largely token-distribution based; richer internal representations (e.g., learned probes on hidden states) may yield substantially higher predictiveness but would change the compute/engineering trade-offs. We also evaluate primarily under deterministic decoding ($T = 0$); future work should assess how telemetry behaves under stochastic decoding and long-form generation. Finally, we evaluated distribution-shift signals such as ΔL_1 and found them generally weaker than entropy/surprisal for early error prediction in our settings; we include an AUC-split analysis and a small top- k ablation in Appendix A.

8 Conclusion and Outlook

We introduced an “EEG-like” telemetry layer for LLM decoding and showed that early-window uncertainty dynamics differ across models and tasks and predict failures above random on labeled tasks. We further demonstrated that telemetry can drive simple downstream decisions such as retry and routing, providing actionable value without parsing CoT text.

We view this work as a first step toward systems that progressively reduce reliance on human-token interfaces, including latent-only reasoning loops (where token generation is reserved for final communication) and latent-to-latent model communication as a compact “meta-language” for machine-to-machine interaction.

A Addendum: AUC Split, Top- k Drift, and k -Ablation

Two reviewer-facing details are easy to miss when aggregating “incorrect” labels and truncating distributions: (i) many “failures” are *format/interface* violations rather than semantic errors, and (ii) top- k drift metrics depend on how the truncated supports are aligned across steps.

AUC split (format vs semantic). Table 3 decomposes AUC into (a) predicting strict format failures, and (b) predicting semantic wrong answers *conditioned on format correctness*. This split is computed *post hoc* from logged telemetry and labels (no model re-run). On GSM8K, early entropy/surprisal are moderately predictive for both format and semantic failures. On TriviaQA, format failures dominate (722/1000), but the strongest semantic predictiveness emerges *within the format-ok subset* (AUC $\approx$ 0.84–0.87).

Table 3: AUC split on canonical full base-only runs. “early” uses the same early-window features as in the main results (entropy_norm_early, surprisal_early).

Task	Failure type	n (pos)	Entropy AUC (early)	Surprisal AUC (early)
GSM8K ($n = 1319$)	format_fail	1319 (184)	0.699	0.713
GSM8K ($n = 1319$)	semantic_wrong format_ok	1135 (461)	0.609	0.601
TriviaQA ($n = 1000$)	format_fail	1000 (722)	0.419	0.361
TriviaQA ($n = 1000$)	semantic_wrong format_ok	278 (195)	0.871	0.841

Token-aligned top- k drift. When computing $\Delta_t = \|p_t^{(k)} - p_{t-1}^{(k)}\|_1$ on truncated top- k distributions, we align by *token ID* on the union support $\text{top-}k(t) \cup \text{top-}k(t-1)$ and treat missing mass as zero. This definition is used throughout this paper.

k -ablation. Table 4 shows that AUC varies only weakly as k increases from 20 to 200 in a GSM8K base-only run, suggesting that top- k truncation is not a dominant source of variance in our setting.

Table 4: Top- k ablation on GSM8K (base-only, $n = 200$). We report AUC for format failures and for semantic wrong answers conditioned on format correctness.

Top- k	AUC _{format} (Ent)	AUC _{sem fmt} (Ent)	AUC _{format} (Spr)	AUC _{sem fmt} (Spr)
20	0.768	0.641	0.770	0.638
50	0.774	0.658	0.769	0.647
200	0.775	0.659	0.769	0.649

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